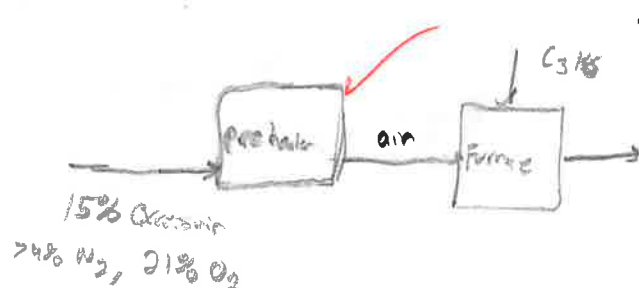


Propane (C_3H_8) is to be burned with 15% excess air. Before entering the furnace, the air is preheated from $0^\circ C$ to $300^\circ C$. If the feed rate of propane is 3.82×10^6 L(STP)/hr, at what rate must heat be transferred (in kJ/hr) to the air in the preheater? Assume complete combustion. Data is on the back of this sheet. At STP ($0^\circ C$ and 1 atm) an ideal gas has a molar volume of 22.4 L/mole.



$$3.82 \times 10^6 \text{ L/hr}$$

$$\hat{H}_{air}(0^\circ C) = -0.72 \text{ kJ/mol}$$

$$\hat{H}_{air}(300^\circ C) = 8.17 \text{ kJ/mol}$$



$$\dot{Q} = n \Delta h$$

$$\frac{P(\%)}{T} = R = \frac{P_{STP} V_{STP}}{T_{STP}}$$

$$(3.82 \times 10^6 \text{ L/hr}) \left(\frac{1 \text{ mole}}{22.4 \text{ L}} \right) = 170,536 \text{ mol } C_3H_8/\text{hr}$$

$$(170,536 \text{ mol } C_3H_8/\text{hr}) \left(\frac{5 \text{ mol } O_2}{\text{mol } C_3H_8} \right) = 852,674 \text{ mol } O_2/\text{hr}$$

$$(852,674) \left(\frac{1}{0.21} \right) (1.15 \text{ (15% excess)}) = 4,664,430 \text{ mol air/hr}$$

$$\dot{Q} = (4,664,430 \text{ mol air/hr}) \cdot (8.17 - (-0.72)) \text{ kJ/mol air}$$

$$= 41,511,235 \text{ kJ/hr}$$

Table B.8 Specific Enthalpies of Selected Gases: SI Units

$\hat{h}(\text{kJ/mol})$							
Reference state: Gas, $P_{\text{ref}} = 1 \text{ atm}$, $T_{\text{ref}} = 25^\circ\text{C}$							
T	Air	O_2	N_2	H_2	CO	CO_2	H_2O
0	-0.72	-0.73	-0.73	-0.72	-0.73	-0.92	-0.84
25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	2.19	2.24	2.19	2.16	2.19	2.90	2.54
200	5.15	5.31	5.13	5.06	5.16	7.08	6.01
300	8.17	8.47	8.12	7.96	8.17	11.58	9.57
400	11.24	11.72	11.15	10.89	11.25	16.35	13.23
500	14.37	15.03	14.24	13.83	14.38	21.34	17.01
600	17.55	18.41	17.39	16.81	17.57	26.53	20.91
700	20.80	21.86	20.59	19.81	20.82	31.88	24.92
800	24.10	25.35	23.86	22.85	24.13	37.36	29.05
900	27.46	28.89	27.19	25.93	27.49	42.94	33.32
1000	30.86	32.47	30.56	29.04	30.91	48.60	37.69
1100	34.31	36.07	33.99	32.19	34.37	54.33	42.18
1200	37.81	39.70	37.46	35.39	37.87	60.14	46.78
1300	41.34	43.38	40.97	38.62	41.40	65.98	51.47
1400	44.89	47.07	44.51	41.90	44.95	71.89	56.25
1500	48.45	50.77	48.06	45.22	48.51	77.84	61.09